

Adding New Creators to Final Evolution Lab

This guide explains how to add new creators to the Creator Card System.

Quick Start

```
# 1. Copy the template
cp Tools/creator_template.json Tools/my_new_creator.json

# 2. Edit the template with creator info
nano Tools/my_new_creator.json

# 3. Validate the data
python3 Tools/add_creator.py --validate Tools/my_new_creator.json

# 4. Add to the database
python3 Tools/add_creator.py --file Tools/my_new_creator.json

# 5. Generate visual assets
python3 -m scripts.creator_pipeline.generate_creator_assets --creator new_creator_id

# 6. (Optional) Generate 3D model
python3 -m scripts.creator_pipeline.meshy_creator_model --creator new_creator_id

# 7. Import into UE5
# Run EditorPython/fel_import_creator_assets.py in UE5 Editor
```

Interactive Mode

For a guided setup:

```
python3 Tools/add_creator.py --interactive
```

Creator Template Fields

Required Fields

Field	Type	Description	Example
<code>id</code>	string	Unique lowercase ID	<code>"john_doe"</code>
<code>display_name</code>	string	Full display name	<code>"John Doe"</code>
<code>tagline</code>	string	Short catchphrase (max 50 chars)	<code>"The Sharpshooter"</code>
<code>bio</code>	string	Full biography (100-500 chars)	<code>"Professional basketball..."</code>
<code>location</code>	string	Home location	<code>"Venice Beach, CA"</code>
<code>primary_sport</code>	string	Main sport	<code>"Basketball"</code>
<code>position</code>	string	Playing position	<code>"Guard"</code>
<code>tier</code>	enum	Creator tier	<code>"Community"</code>
<code>card_style</code>	enum	Card visual style	<code>"Standard"</code>

Optional Fields

Field	Type	Default	Description
<code>height</code>	string	<code>""</code>	Display height
<code>weight</code>	string	<code>""</code>	Display weight
<code>has_3d_model</code>	bool	<code>false</code>	Whether a 3D model exists
<code>is_featured</code>	bool	<code>false</code>	Show in featured carousel
<code>is_playable</code>	bool	<code>false</code>	Can players play as this creator
<code>sort_order</code>	int	99	Gallery display order

Tier Values

- **Featured** — Top-tier creators with full content (Elijah Bonds, Amir Smith)
- **Verified** — Verified creators with good content (Eric Nash)
- **Community** — Community contributors

- **ComingSoon** — Placeholder for upcoming creators

Card Style Values

- **Standard** — Default card appearance
- **Premium** — Enhanced visuals with subtle effects
- **Legendary** — Gold/animated card border
- **Holographic** — 3D holographic effect (for creators with 3D models)

Content Requirements

Images

Asset	Size	Format	Notes
Profile Image	512×512 px	PNG	Circular crop in UI
Banner Image	1920×400 px	PNG	Wide landscape
Card Background	400×530 px	PNG	Portrait orientation

Videos

Format	Codec	Resolution	Max Duration
MP4	H.264	1080p or 720p	120 seconds

3D Models

Format	Polygons	Textures
GLB/FBX	<50,000	PBR (4K max)

Highlight Types

- `gameplay` — Game/match footage
- `dunk` — Dunk highlights
- `training` — Training/coaching content
- `magic_reveal` — AR magic reveal videos
- `3d_model` — 3D model showcase (no video, model viewer)

Signature Move Categories

- `dunk` — Dunks and slam moves

- `shooting` — Shooting techniques
 - `ball_handling` — Dribble moves and crossovers
 - `defense` — Defensive moves
 - `passing` — Passing techniques
 - `post_move` — Post-up moves
 - `transition` — Fast break / transition plays
-

Generating Visual Assets

The asset pipeline uses **Luma AI** for images and **Meshy** for 3D models.

Prerequisites

```
# Set API keys in .env
LUMA_API_KEY=your_key_here
MESHY_API_KEY=your_key_here
```

Generate Profile Images & Banners

```
# Single creator
python3 -m scripts.creator_pipeline.generate_creator_assets --creator john_doe

# All creators
python3 -m scripts.creator_pipeline.generate_creator_assets --all

# Icons only
python3 -m scripts.creator_pipeline.generate_creator_assets --icons-only

# Dry run (preview without generating)
python3 -m scripts.creator_pipeline.generate_creator_assets --all --dry-run
```

Generate 3D Model (via Meshy)

```
# Add custom prompt to CREATOR_MODEL_PROMPTS in meshy_creator_model.py first
python3 -m scripts.creator_pipeline.meshy_creator_model --creator john_doe

# List recent Meshy tasks
python3 -m scripts.creator_pipeline.meshy_creator_model --list-tasks
```

Importing into Unreal Engine 5

Automatic Import

1. Open the UE5 project
2. Go to **Tools > Execute Python Script**
3. Select `EditorPython/fel_import_creator_assets.py`
4. The script will import all found creator assets

Manual Import

1. Open **Content Browser**
2. Navigate to `/Game/FEL/CreatorAssets/`
3. Import images from `GeneratedAssets/CreatorAssets/`
4. Import models from `GeneratedAssets/CreatorModels/`

Asset Directory Structure

```
Content/FEL/
├── CreatorAssets/
│   ├── Profiles/           # 512x512 profile images
│   ├── Banners/           # 1920x400 banners
│   ├── Backgrounds/       # Card backgrounds
│   ├── Thumbnails/        # Video thumbnails
│   └── Icons/              # Stat icons, badges
├── CreatorModels/
│   └── AmirSmith/         # Amir's 3D model
├── Videos/
│   ├── MagicReveal/       # Magic reveal videos
│   ├── ElijahBonds/       # Elijah's highlights
│   ├── AmirSmith/        # Amir's highlights
│   └── EricNash/          # Eric's highlights
└── Data/
    └── CreatorProfiles.json
```

C++ Integration

Access Creator Data in Code

```
#include "FELCreatorManager.h"

// Get the manager
UFELCreatorManager* CM = GetGameInstance()->GetSubsystem<UFELCreatorManager>();

// Get a profile
FCreatorProfile Profile;
if (CM->GetCreatorProfile(TEXT("elijah_bonds"), Profile))
{
    UE_LOG(LogTemp, Log, TEXT("Creator: %s"), *Profile.DisplayName);
}

// Get featured creators
TArray<FCreatorProfile> Featured = CM->GetFeaturedCreators();

// Select a creator
CM->SelectCreator(TEXT("amir_smith"));

// Load 3D model
USkeletalMesh* Model = CM->LoadCreatorModel(TEXT("amir_smith"));
```

Challenge Integration

```
#include "FELCreatorChallengeManager.h"

// Add to game mode actor
UFELCreatorChallengeManager* Challenges = NewObject<UFELCreatorChallengeManager>(this)
;
Challenges->ActivateCreatorChallenges(TEXT("elijah_bonds"));

// Report progress
Challenges->IncrementScore(TEXT("eb_dunk_gauntlet"), 1);
Challenges->ReportScore(TEXT("eb_500_shot_challenge"), CurrentShots);
```

Blueprint Integration

Creator Gallery

- 1. Create a Blueprint child of `UFELCreatorGalleryWidget`
- 2. Design the UMG layout in the Widget Designer
- 3. Bind widgets using the `BindWidget` names from the C++ class
- 4. Call `InitializeGallery()` to populate

Creator Profile Page

- 1. Create a Blueprint child of `UFELCreatorProfileWidget`
- 2. Design the full profile layout
- 3. Call `LoadProfile(CreatorID)` to display a creator

Main Menu Integration

- 1. Add a “Creators” button to the main menu
- 2. On click, create and display the Creator Gallery widget
- 3. The gallery handles navigation to individual profiles

Current Creators

Creator	Tier	3D Model	Highlights	Challenges
Elijah Bonds	Featured/Le-gendary	No	10 (7 Magic Re-veal + 3 Dunks)	3
Amir Smith	Featured/Holo-graphic	Yes (Meshy)	3	2
Eric Nash	Verified/Premi-um	No	3	3

Troubleshooting

“CreatorProfiles.json not found”

- Ensure the file is at `Content/FEL/Data/CreatorProfiles.json`
- Run `python3 Tools/add_creator.py --list` to verify

“Failed to load 3D model”

- Run the import script in UE5 Editor first
- Check that GLB/FBX files exist in `GeneratedAssets/CreatorModels/`

“API key not set”

- Check `.env` file for `LUMA_API_KEY` and `MESHY_API_KEY`
- The asset generator creates placeholder entries if no key is set

Adding to the Meshy pipeline

1. Edit `scripts/creator_pipeline/meshy_creator_model.py`
2. Add a new entry to `CREATOR_MODEL_PROMPTS`
3. Run `python3 -m scripts.creator_pipeline.meshy_creator_model --creator new_id`